

Network Security: Llareggub Security Assessment - Ross Greenall 22503533

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Introduction

LLareggub has a multi-site network topology which includes Internet access, servers, wireless routers and IoT devices. This report contains a thorough assessment of the current network security infrastructure based on the Packet Tracer configuration file provided by the CEO. Additionally, this document provides security recommendations that can be applied through both hardware and software updates which will help mitigate security incidents in the future.

Server Role

Llareggub hosts four centralised servers that provide core business functionality and communication. These servers are the Finance Server, Sales Server, Email Server and a Web Server which provides DNS services. These systems are critical assets in the business because they store all department data and services relied on by all devices within the WAN.

The Web Server is particularly important because DNS provides name resolution for internal requests and Internet requests, making it a primary dependency for network connectivity. Therefore, any compromise to this infrastructure will lead to a loss in core business functionality and the availability of service.

Wireless Access Points & IoT Role

The Wireless AP within the topology provide wireless communication with the business network [5], this is crucial for office work and IoT devices. Currently, several AP in three regional sites allow users to connect and communicate, especially where cabling and wired connection is not physically possible for business use. These AP are critical pieces of infrastructure for the business as they allow the communication between sites and the central headquarters.

Wireless connectivity increases the flexibility of the business and allows for the extension of the network, however this also widens the attack surface of the network. Multiple APs in several locations represent critical infrastructure for both availability and security, if wireless connectivity is disrupted then functionality will be reduced at the affected sites.

The IoT devices located at the Stores and Production site are critical for maintaining specific temperature and humidity controls, which if affected can cause damage to stored or produced goods. As such, they operate continuously and any disruption could cause huge damage to business production.

Vulnerability Assessment, Impact Analysis & Recommendations

Wide Area Network (WAN) Vulnerabilities

Untrusted WAN Traffic

Currently within the topology, a Cisco 2911 router is acting as the Internet for the purposes of demonstrating a wide-area network (WAN) in the lab environment, the Internet is not a trusted source of traffic [1], and any incoming traffic originating from the WAN is potentially malicious and logged for investigation.

Edge Router Policy Enforcement

As stated in the Firewall Policy, the security protocols and policies for inbound traffic should be enforced at each site router, creating a security boundary for the individual locations, utilising a deny all default approach [6]. This is a perfectly acceptable design structure for the network, however in the current configuration, inbound traffic from the WAN is trusted at all sites, posing a significant increase in the likelihood of lateral movement between sites.

Impact Analysis

Device(s)	Vulnerability	Systems that could be impacted?	How it would impact operations	CIA Impact	Business Severity	Policy
WAN 1	Untrusted WAN topology not implemented.	All sites, HQ servers and user endpoints.	Compromise on one site can result in others being affected, potential access to data and services. Entire business	Confidentiality, Integrity.	Critical	Firewall Policy

			operations can be disrupted.			
WAN 1	OSPF without Authentication.	Routing between all routers.	Route altering could cause traffic outages, interception and manipulation of data in transfer. Multiple site disruption.	Confidentiality, Availability and Integrity	Critical	Authentication Policy

Recommendations

All traffic crossing the WAN should be treated as untrusted [1], even when it is exchanged between Llareggub's own sites. Although the WAN is used to support normal business communication, it still represents an exposed layer between separate locations and should therefore not be relied upon as a trusted means of communication. Without additional protection, cross-site traffic may be vulnerable to interception and spoofing [1].

The most important improvement would be to protect cross-site communication using encrypted tunnels such as IPsec. This would ensure that data travelling between the Head Office, Sales offices and the Stores and Production site is protected in terms of confidentiality and integrity. Encrypting WAN traffic would reduce the risk of attackers viewing sensitive information in transit or tampering with communications between sites.

A second improvement would be to apply stricter traffic filtering at the edge routers [1]. By allowing only the specific services required for business operation, such as approved web, DNS, and email, should be allowed across the WAN. This would significantly reduce the attack surface and prevent unnecessary or unauthorised traffic from moving between locations.

Routing security should also be strengthened [2]. As the WAN relies on cross-site routing to maintain connectivity, the routing process itself must be protected from alteration. Enabling OSPF authentication would reduce the likelihood of false routes being created within the network.

Edge Router Vulnerabilities

As the WAN is to be treated as untrustworthy, each site router must function as a security boundary between the internal network and WAN traffic. This section assesses whether the edge routers at each location can implement effective control measures to maintain the security design suggested in the previous section.

Observed Vulnerabilities

Lateral Movement and Traffic Restriction

As previously stated, the implementation of the Firewall Policy and Authentication policy is required for the creation of boundaries between the edge routers on the WAN. In the current topology the edge routers are not configured to implement this restrictive access policy at the site edge, making them susceptible to malicious traffic and allowing the possibility of lateral movement between site networks. The fundamental design structure of the network topology requires firewall restrictions to exist at external facing nodes, and the most practical point of implementation is at each edge router, to mitigate the issues mention and begin building the foundations for additional security measures.

Lack of Management Configuration Authentication

Furthermore, in the current topology, the edge routers are not secured by any management configuration. With a clear lack of encrypted administrative access such as Secure Shell (SSH), a rogue employee could gain console access to the routers and undermine any security features that are implemented in future. Although connectivity is not possible from any client in the network, the routers have Telnet remote access enabled but there is not a password set, with proper access an attacker could enable a password and take control of the router and configure it in any way to benefit their intentions.

Impact Analysis

Device(s)	Vulnerability	Systems that could be impacted?	How it would impact operations	CIA Impact	Business Severity	Policy
2811 Head Office Manchester, 2811 Remote Sales North, 2811 Remote Sales South, 2811 Stores and Production	Edge router without boundary restrictions on incoming traffic.	All Edge Routers, potentially stemming to internal network devices and infrastructure.	Any untrusted traffic from the WAN is accepted without proper filtering or monitoring. Lack of control of traffic flow can disrupt confidentiality, availability and integrity of all business communication and internet usage.	Confidentiality, Integrity.	High	Firewall Policy
2811 Head Office Manchester, 2811 Remote Sales North, 2811 Remote Sales South, 2811 Stores and Production	Insecure remote access to router configuration.	All Edge Routers, potentially stemming to internal network devices and infrastructure.	No implementation of SSH, lack of defined password within running configuration file. Any malicious actor with access to the console can tamper with router configuration. Complete control over network to a threat actor, no ability to remotely manage device function or security.	Confidentiality, Availability and Integrity	Critical	Firewall Policy, Remote Access Policy, Authentication Policy

Recommendations

To begin to rectify the security problems that exist within the edge routers, the first thing that must be done is to limit all access to authorised personnel only, namely system administrators. This can be achieved through physical security, by placing the edge routers in a secure location which is only accessible if explicitly authorised by the head of the Security Operations Centre, and relying on configuration by encrypted remote access protocols such as SSH [4], with strong administrator credentials and MFA as noted in the Remote Access Policy. This mitigates the risk of a rogue employee taking control of edge routers and undermining future security improvements. Further to the implementation of SSH, unsuccessful access attempts must be logged and investigated in a timely manner.

Secondly, to ensure the routers adhere to the proposed control design and deny all approach [6], each router must be configured to include the previously mentioned OSPF authentication, this ensures that adjacent connections are only accessible to authenticated users, which bolsters the untrusted nature of traffic from the WAN. Each edge router must also be configured with access-list controls to permit only the minimum amount of network traffic that is necessary, primarily services and administration functionality.

Network Switch Vulnerabilities

Observed Vulnerabilities

Lack of Internal Segmentation

Within the current design there is only one VLAN in use, VLAN 1, with no separation between users, servers, wireless connectivity and any other devices. This flat VLAN design increases exposure to lateral movement following endpoint failure, an attacker has the potential to communicate with numerous internal assets without any traversal. Without internal segmentation using VLANs, there is currently no structure where access is dependent on user permissions, ideally administrators should have access to all internal departments, and standard users only having access to the required services needed for business operation.

Daisy Chain Switch Configuration

The daisy chained topology of the switch structure also introduces dependency and monitoring issues. Switch0 is a SPOF for all devices on the network, meaning an attacker attempting to disrupt availability has a direct target, additionally, backup edge routers should also be introduced as each of these also act as SPOFs for each location. Furthermore, the reliance on a singular switch means a higher concentration of traffic at this point, leading to slower response times during peak traffic, even more so during possible DoS attacks.

Dynamic Trunk Negotiation

Currently, Fa0/1 is trunking on Switch1 and Switch0 using VLAN 1, it is currently set to desirable mode using DTP. This is a potential security risk as when future VLANs are implemented, VLAN traffic could be carried onto a device that was not trusted, therefore these trunks should be explicitly defined and not assigned dynamically using DTP.

```
Port      Name      Status      Vlan      Duplex  Speed Type
Fa0/1     connected connected    trunk     a-full  a-100 10/100BaseTX
```

Open Non-connected Switch Ports

All three switches have many ports which are not in use in the current network topology, by default these ports are not connected but are still enabled, these are potential access points to the LAN as a network device can be connected at any time and the port can be open for communication, an attacker could gain physical access to the switch and begin altering trunk behaviour or VLAN structure affecting all aspects of the CIA triad.

```
Switch#show interfaces status
Port      Name      Status      Vlan      Duplex  Speed Type
Fa0/1     connected connected    trunk     a-full  a-100 10/100BaseTX
Fa0/2     connected connected    1         a-full  a-100 10/100BaseTX
Fa0/3     notconnect notconnect  1         auto    auto  10/100BaseTX
Fa0/4     notconnect notconnect  1         auto    auto  10/100BaseTX
Fa0/5     notconnect notconnect  1         auto    auto  10/100BaseTX
Fa0/6     notconnect notconnect  1         auto    auto  10/100BaseTX
Fa0/7     notconnect notconnect  1         auto    auto  10/100BaseTX
Fa0/8     notconnect notconnect  1         auto    auto  10/100BaseTX
Fa0/9     notconnect notconnect  1         auto    auto  10/100BaseTX
Fa0/10    notconnect notconnect  1         auto    auto  10/100BaseTX
Fa0/11    notconnect notconnect  1         auto    auto  10/100BaseTX
Fa0/12    notconnect notconnect  1         auto    auto  10/100BaseTX
Fa0/13    notconnect notconnect  1         auto    auto  10/100BaseTX
Fa0/14    notconnect notconnect  1         auto    auto  10/100BaseTX
Fa0/15    notconnect notconnect  1         auto    auto  10/100BaseTX
Fa0/16    notconnect notconnect  1         auto    auto  10/100BaseTX
Fa0/17    notconnect notconnect  1         auto    auto  10/100BaseTX
Fa0/18    notconnect notconnect  1         auto    auto  10/100BaseTX
Fa0/19    notconnect notconnect  1         auto    auto  10/100BaseTX
Fa0/20    notconnect notconnect  1         auto    auto  10/100BaseTX
Fa0/21    notconnect notconnect  1         auto    auto  10/100BaseTX
--More--
```

Impact Analysis

Device(s)	Vulnerability	Systems that could be impacted?	How it would impact operations	CIA Impact	Business Severity	Policy
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Switch0, Switch3, Switch1	Singular VLAN implementation, no internal segmentation of devices.	Manchester Head Office internal network and functionality of other sites with server infrastructure.	Currently there is no restriction in communication between different areas of the LAN, allowing for easy lateral movement through the network, also no designation between user and management permissions on the network for access.	Confidentiality, Integrity.	Medium	Firewall Policy
Switch0, Switch5	Daisy Chained Network Topology SPOF	Manchester Head Office internal network.	Any disruption to the functionality of these switches will cause all devices within the LAN to be unable to communicate, rendering the network in this location useless. Significant loss of business functionality in the event that these switches fail, resulting in complete outage of the Head Office.	Availability	Low	No policy for SPOF.
Switch1, Switch0	Dynamic Trunk Negotiation Enabled	Manchester Head Office internal network.	Untrusted network devices can be asked to trunk allowing for easy traversal through the network topology. Could allow an attacker with physical access to the network switches control over the network configuration.	Integrity	Medium	Authentication Policy, Remote Access Policy
Switch0, Switch3, Switch1	Open non-connected ports	Manchester Head Office internal network.	Any access to the physical switch will allow an attacker to	Integrity	Low	Authentication Policy

			add devices to the network. Not a major operational impact as the case requires physical access.			
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Recommendations

The switching configuration at the head office should be redesigned to support internal segmentation rather than wide connectivity through VLAN 1. At present, a flat switching model increases the likelihood that a compromise in one part of the network will spread laterally into other departments and services. Departmental VLANs should therefore be introduced for Finance, Sales, IT Support, Servers, Wireless and IoT-connected systems.

Inter-VLAN communication should not be permitted by default. Instead, traffic between segments should be explicitly controlled using ACLs implemented at the boundary routers.

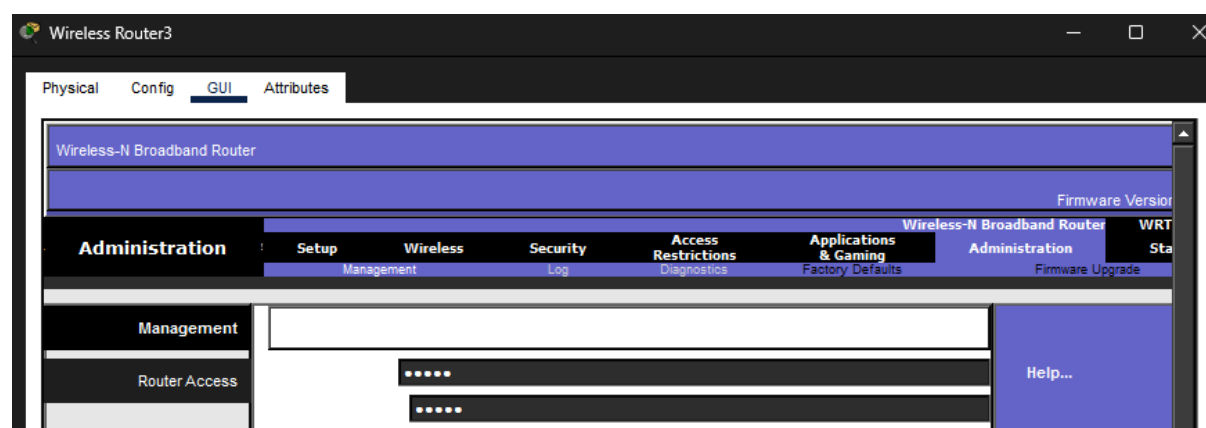
Switch hardening should also be improved [2]. Unused ports should be administratively shut down, port security should be enabled to restrict unauthorised device connections, and trunk links should be limited to only the VLANs that are explicitly required. The current broad trunk allowance creates unnecessary exposure and increases the likelihood of misconfiguration. Disabling DTP where it is not needed would further reduce the risk of unauthorised trunk negotiation.

Wireless Access Point Vulnerabilities

Observed Vulnerabilities

Access Points Increase Attack Surface & Poor Credentials

The APs at several locations within the infrastructure provide the additional wireless connectivity that the business requires to operate but they also increase the attack surface of the business. The primary concern for these access points is the potential access to the internal LAN at these sites. This is more likely to occur in scenarios where poor security practices such as improper authentication configuration is implemented. In the network topology, the APs have short and common password schemes which renders the authentication useless against even rudimentary brute force attacks. Furthermore, there exists no access restrictions regarding application types or users, allowing for the communication of any requests over the internal LAN.



No Segmentation of Wireless Network

Within the internal network at these sites, there exists no segmentation between user types. Especially so on the APs at the remote locations. This lack of segmentation between users, guests, management and other devices allows for easy traversal and targeted attacks across the wider network model and critical business infrastructure, even from

lower trust devices. This poses a significant risk to the operation of the business and the use of VLANs and Policies for trusted devices should be enforced, particularly so at these insecure APs, with a specific focus on traffic logging and investigation as these can often be overlooked in network assessments.

Consumer Wireless Access Points

The implemented APs are not sufficient enough for the scale and importance of the network. The WRT300N is a legacy device and as such does not implement many modern security patches that have been implemented since its release in 2007. Furthermore, the capable speed of these APs is significantly slower than WiFi 6 standards and could lead to bottlenecks and reduced system capacity as the business grows, or an attacker attempts a denial of service attack by exploiting previously mentioned vulnerabilities in credentials. The positive points are that in the current configuration WPS, remote authentication and UPnP are currently disabled, which eliminates several vulnerabilities from the device.

IoT Reliance on APs

The current topology indicates that the IoT devices located in the Stores and Production facility, are all connected to a single AP. This implementation could cause significant outages, particularly as many IoT devices suffer from critical unpatched security vulnerabilities that are unknown to most users. This combined with the singular AP for all continuously connected IoT devices, could result in severe business disruption in the event of an attack.

No Centralised Access Point Controller

Currently all access points are decentralised and configured at the individual sites, this poses a security risk as credentials may be mismanaged and the configuration of APs may not meet the standards required by the security policies. This could result in the loss of network connectivity and increase the attack surface of the business.

Impact Analysis

Device(s)	Vulnerability	Systems that could be impacted?	How it would impact operations	CIA Impact	Business Severity	Policy
WR0, WR1, WR2, WR3	Poor/Low quality credentials used for admin controls.	IoT Devices at Stores and Production, Wireless devices and wider network.	Threat actors could take control of the wireless connectivity and functionality of the network, this will also impact the function of IoT devices present at the Stores and Production department causing prominent business disruption.	Integrity	High	Wireless Network Security Policy, Authentication Policy
WR0, WR1, WR2, WR3	Lack of internal segmentation on wireless networks.	Wireless AP, IoT devices.	There is a lack of segmentation within the wireless networks with no specific roles or separation between guests, administrators, users and IoT devices, therefore any	Integrity	Medium	Remote Access Policy, Authentication Policy, Wireless Network Security Policy

			device connected to an AP within the topology is free to communicate with all other devices.			
IoT Devices	Over reliance on APs for IoT devices	IoT Devices	SPOF for IoT devices connected to wireless AP, if the AP is compromised or loses connection to the Stores and Production router, all IoT devices will be unable to communicate and stock could be damaged.	Availability	Medium	Wireless Network Security Policy
WR0, WR1, WR2, WR3	Consumer grade APs used in commercial setting.	All remote sites using APs	The type of router used in these locations is a WRT3000N, which is a consumer grade AP not suitable for larger commercial scale usage. These routers have less built in security systems and as such leave critical business assets less secured than required.	Availability	Low	Wireless Network Security Policy
WR0, WR1, WR2, WR3	No centralised Wireless Access Controller	All networks utilising wireless access.	Network credentials and configurations for the APs in all locations are managed separately, this could cause inconsistencies within security practice, leading to an increase in the attack surface of the business.	Confidentiality, Integrity, Availability	Low	Wireless Network Security Policy, Authentication Policy

Recommendations

Wireless access should be treated as a high-risk entry point because it extends the internal network beyond physically controlled infrastructure. The first priority should be to replace weak and easily guessable wireless credentials with

strong passphrases. This would reduce the likelihood of brute-force attacks.

A second improvement would be to separate wireless traffic from critical internal services by placing wireless APs into dedicated VLANs. Staff wireless traffic should only be permitted to access the internal resources necessary for business operation, while IoT devices and any visitor connectivity should be isolated into separate segments. This would reduce the risk of lateral movement if a wireless device were compromised.

The current use of consumer-grade wireless routers in a business environment should also be addressed. Commercial-grade access points would provide stronger security features, more consistent policy enforcement and better monitoring capability.

In addition, wireless security should be strengthened through regular password review, disabling unnecessary management interfaces, changing all default administrative settings, and restricting management access to authorised IT systems only.

Software Vulnerabilities

FTP and TFTP Lacking Credentials

Currently on the Finance, Sales, Email and Web Servers, FTP and TFTP are enabled. TFTP is primarily used for quick data transfer over UDP, which is acceptable for security updates. FTP, which uses TCP, is the service used by the business for the distribution of files, the issue lies in that both of these services are enabled on all servers without any credential requirements. Any attacker that gains access to the network can distribute malicious files and take over the businesses core network infrastructure.

IoT No Authentication Enabled

All the IoT devices within the Stores and Production location have wireless connectivity enabled, this allows for the connection of IoT devices to the wider LAN network. This gives the business the ability to manage the IoT devices without physical interaction if they are constrained by location or placement. The primary risk factor associated with IoT devices is insecure network configuration. In its current state, all the IoT devices have no form of authentication enabled for wireless connections, leaving them vulnerable to manipulation.

Impact Analysis

Device(s)	Vulnerability	Systems that could be impacted?	How it would impact operations	CIA Impact	Business Severity	Policy
All Servers	FTP/TFTP Credentials Missing	All critical file access infrastructure and data in storage.	Without proper authentication systems in place for the FTP/TFTP protocols, an attacker could upload, download and alter files within the businesses server, causing significant disruption to business function.	Confidentiality, Integrity, Availability	Severe	Authentication Policy, Remote Access Policy
IoT Devices	No credentials on IoT devices.	IoT and the wider business product distribution line.	IoT devices are accessible and able to be altered by malicious actors, causing significant and	Availability	Medium	Remote Access Policy, Authentication Policy

			prominent loss of control over the businesses Stores and Production line.		
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Recommendations

The use of FTP and TFTP without authentication presents a risk to core business servers. Where file transfer functionality is still needed, it should be replaced with authenticated and encrypted alternatives such as SFTP or SCP. This would reduce the risk of unauthorised file access, malicious file uploads and the interception of data in transit.

Where legacy services must remain enabled temporarily, access should be restricted through ACLs, limited to specific management devices, and monitored [3]. This reduces exposure while a more secure replacement is put in place. The organisation should also review which servers actually require file transfer services, as enabling unnecessary services across finance, sales, email and web systems increases the attack surface.

More broadly, the organisation should implement a formal hardening policy for network devices [2]. Servers and embedded devices should be reviewed regularly to ensure that unnecessary services are disabled, default settings are configured properly and security updates are applied in a timely manner. This would improve the overall resilience of the network against opportunistic attacks.

Executive Summary & List of Priorities

Priority	Vulnerability	Risk to Business	Recommended Control	Security Benefit
1	Untrusted WAN traffic	Interception, spoofing, route abuse, cross-site compromise	VPN, ACLs, OSPF authentication	Protects confidentiality and integrity of cross-site traffic
2	Flat internal network / poor segmentation	Lateral movement to critical systems	VLAN segmentation and inter-VLAN ACLs	Limits attacker movement and reduces damage radius
3	Weak router and admin security	Device takeover, traffic redirection, policy bypass	SSH only, strong authentication, restricted admin access	Protects management operations
4	FTP/TFTP without authentication	File tampering, malicious uploads, server compromise	Disable unnecessary services, use SFTP/SCP	Reduces data manipulation
5	Weak wireless controls	Unauthorised LAN access	Strong credentials, business-grade APs, wireless segmentation	Reduces attack surface at remote sites
6	IoT without authentication	Device manipulation and access into LAN	Isolated VLAN, authentication, restricted flows	Prevents low-trust devices from accessing core business systems
7	Poor monitoring	Delayed detection of incidents	IDS/IPS	Improves detection and response

References

[1] S. Rose, O. Borchert, S. Mitchell and S. Connelly, *Zero Trust Architecture*, Gaithersburg, MD, USA: National Institute of Standards and Technology, NIST SP 800-207, 2020. doi: 10.6028/NIST.SP.800-207.

[2] Joint Task Force, *Security and Privacy Controls for Information Systems and Organizations*, Gaithersburg, MD, USA: National Institute of Standards and Technology, NIST SP 800-53 Rev. 5, 2020. doi: 10.6028/NIST.SP.800-53r5.

[3] M. Souppaya and K. Scarfone, *Guide to Enterprise Patch Management Planning*, Gaithersburg, MD, USA: National Institute of Standards and Technology, NIST SP 800-40 Rev. 4, 2022. doi: 10.6028/NIST.SP.800-40r4.

- [4] T. Ylonen and C. Lonvick, *Security of Interactive and Automated Access Management Using Secure Shell (SSH)*, Gaithersburg, MD, USA: National Institute of Standards and Technology, NISTIR 7966, 2015.
- [5] J. F. Kurose and K. W. Ross, *Computer Networking: A Top-Down Approach*, 8th ed. Harlow, U.K.: Pearson, 2021.
- [6] J. Garbis, *Zero Trust Security: An Enterprise Guide*. Berkeley, CA, USA: Apress, 2021